

[illegible]

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

COMMON TO B.E –CSE & M.TECH - CSE

Answer ALL questions

Maximum: 70 Marks

(5x 1 = 5 Marks)

- 24-25-10

PART - B

(5x 3 = 15 Marks)

6. **CO1** Recall the equations for finding speedup and efficiency of parallel systems.
7. **CO2** Differentiate between task-parallelism and data-parallelism.
8. **CO2** Outline the semantics of MPI_Send and MPI_Recv.
9. **CO3** Sketch the modern GPU architecture.
10. **CO3** Summarize loop scheduling in OpenMP.

PART - C

(5 x 10 = 50 Marks)

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO2** Build a matrix multiplication program using OpenMP. (10)
12. **CO2** Discuss the Messages and point-to-point communication in MPI. (10)
13. **CO3** Describe shared memory parallel programming with OpenMP. (10)
14. **CO2** Compare multicore, multithreaded and vector processors. (10)
15. **CO2** Discuss Warps and Thread Execution in CUDA. (10)
16. **CO3** Build a CUDA program to perform vector addition and vector multiplication with Blocks and Threads. (10)
17. **CO1** i) Consider the serial run-time is $T_{\text{serial}} = 20$ seconds. Then the run-time of the parallelized part is $0.9 \cdot T_{\text{serial}} / p = 18/p$ and the run-time of the “unparallelized” part is $0.1 \cdot T_{\text{serial}} = 2$. Estimate the speedup and the overall parallel run-time. (5)
CO1 ii) Illustrate the Memory Hierarchies for High Performance Computing. (5)
